Subset Sums in a Fourfold Sumset

A checked continuation from 103 to 100

Sum-of-Subset research project

16 October 2026

Abstract

We continue the repository's independently checked upper bound 103 to 100. The obstruction at size 103 is a possible empty covering unit. We handle this obstruction and the corresponding cases at sizes 102 and 101 by 2,580 explicitly specified linear systems. Freshly generated integer certificates prove every system inconsistent. The supplied verifier uses only the Python standard library and checks all coefficients exactly. A uniform effective-hole test treats the high-hole cases, leaving only one empty unit and, at size 101, at most one outside center. Together with the known construction this gives the checked interval $71 \le N \le 100$. The archived claim $N \le 84$ is stronger and is not certified in its entirety by this work.

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1 Conclusion, conventions, and dependency boundary

For a finite real set A, let

$$\mathrm{FS}(A) = \left\{ \sum_{x \in X} x : X \subseteq A \right\}, \qquad 4A = A + A + A + A.$$

The empty sum belongs to $\mathrm{FS}(A)$, and repetitions are permitted in $4A$. Write $T_r = r(r+1)/2$ and let N be the maximum feasible cardinality.

Theorem 1.1. *If* $\mathrm{FS}(A) \subseteq 4A$, *then* $|A| \le 100$.

This report uses the audited 115 foundation and the structural lemmas proved in the 115-to-103 continuation [1]. In particular it uses that report's corrected peeling argument, positive-side stability and inverse lemmas, common-lattice method, four-extreme estimate, and active-center

identities. It rechecks the size-dependent applicability gates below; it does not merely assume that a statement proved at 104 also holds at 101. The previous theorem $N \le 103$ closes the upper end of the new descent.

The one-empty-unit strategy and its necessary inequalities already appear in the archived upper-86 report [2]. Here we give a compact fresh verification of the required portion, rebuild the complete model universe, regenerate all integer certificates, and replace separate high-hole bootstrap chains by one parameter test. No archived endpoint certificate or historical PASS record is an input to the new verifier. This improves the independently closed bound, not the project's strongest archived numerical claim. It does not determine the exact maximum.

The 71-element example remains

$$[-31, 31]_{\mathbb{Z}} \cup \{\pm 93, \pm 155, \pm 217, \pm 713\}.$$

Its proof and exact certificate are unchanged. Thus the checked range is $71 \le N \le 100$.

2 Reduction to common-lattice scalar parameters

Adjoining zero preserves feasibility: $4A \subseteq 4(A \cup \{0\})$ and $\mathrm{FS}(A \cup \{0\}) = \mathrm{FS}(A)$. For integralization, fix a four-term representation of every subset sum. The resulting homogeneous rational linear system has rational solutions arbitrarily close to its real solution inside its kernel. Preserve the strict coordinate order and the zero coordinate, then clear denominators. Consequently existence can be studied with zero present and with integer coordinates, at each stage of a descending exclusion.

Remove the four most positive and four most negative elements to obtain D, and make two four-extreme peels $D \supset D_1 \supset E$ preserving zero. For $n \in \{101, 102, 103\}$ put

$$d = n - 8, \quad e = n - 16, \quad E = P \cup \{0\} \cup (-Q), \quad a = |P|, \quad b = |Q|, \quad m = a + b = n - 17.$$

The elementary cover and peel comparisons from [1] give

$$|\,\mathrm{FS}(E)| \le 24d + 77.$$

If fewer than eight outer elements were removed, the core is one-sided with at least $n - 16$ nonzero elements. This contradicts $1 + T_{n-16} > 24n + 77$, checked at all three sizes. If the core has p opposite pairs and q unpaired absolute values, the same argument as in [1] gives

$$|\,\mathrm{FS}(E)| \ge \frac{e^2 + 3 + q^2}{4}, \quad q^2 \le 96e + 1073 - e^2, \quad q \equiv e - 1 \pmod 2.$$

The resulting lower bounds on p are 22, 21, 20 at sizes 103, 102, 101. In particular both signs and the required opposite pair are present.

For a positive set X of size r, write

$$\mu(X) = |\,\mathrm{FS}(X)| - 1 - T_r, \quad \delta_0(X) = |\{x \in X : 2x \notin R(X \cup \{0\})\}|, \quad \rho(X) = \mu(X) + 2 - 2\delta_0(X),$$

where $R(X)$ denotes restricted sums of two distinct elements. The inherited estimates are $\rho(X) \ge 0$ for $r \ge 14$, and $\mu(X) \ge r - 3$, $\rho(X) \ge r - 14$ when its support is not a progression.

Set $\Lambda = 1 + T_a + T_b$ and $\Omega = 22d + 97 - \Lambda$. The cover slack is

$$c = 2|2D| + 18d + 85 - |\,\mathrm{FS}(D)| \ge 0.$$

With $s = |\,\mathrm{FS}(E)|$ and $\mu = \mu(P) + \mu(Q)$, let $\kappa = s - \Lambda - \mu$. If ℓ_1, ℓ_2 are the two nonnegative peeling slacks and $g = 4d - 10 - (|R(D)| - |R(D_1)|)$, set $\mathcal{T} = \kappa + \ell_1 + \ell_2 + g$. The inherited exact identity is

$$\rho(P) + \rho(Q) + 2\theta + c + \mathcal{T} = \Omega, \qquad \theta = \delta_0(P) + \delta_0(Q) + 8 - \Delta(D) \ge 0. \tag{1}$$

Here $\Delta(D) = |2D| - |R(D)|$.

The two-set inverse theorem and propagation to D require precisely the strict gates

$$T_{\min(a,b)} - 2 - \Omega > 0, \quad 2m + 4e + 14 - \Lambda + 3\Omega < 0, \quad \Lambda - (4d - 10 + \Omega) > 0. \tag{2}$$

Enumerate both orientations of all a, b at least the paired minimum, with $a + b = m$ and $\Omega \geq 0$. The margins are:

n	Orientations	Paired minimum	First minimum	Second maximum	Third minimum
103	35	22	335	−477	1229
102	36	21	299	−377	1169
101	37	20	264	−277	1109

All hypotheses in the common-lattice argument of [1] are therefore valid. After scaling, D and E are integral. If h_P, h_Q, z, w are the hull-hole counts of $\mathrm{FS}(P), \mathrm{FS}(Q), \mathrm{FS}(E), \mathrm{FS}(D)$ on this lattice, then

$$h_P, h_Q \leq \kappa, \quad z = h_P + h_Q - \kappa, \quad 0 \leq z \leq \min(h_P, h_Q) \leq \kappa, \quad h_X < |\,\mathrm{FS}(X)| - 1, \tag{3}$$

$$w \leq 4z, \qquad \mathcal{T} \geq \max\{z, G - 2z, t_0\}, \quad G = 2d - 12, \quad t_0 = \lceil G/3 \rceil. \tag{4}$$

For $z > 0$ neither side support is a progression, so $\rho(P) + \rho(Q) \geq m - 28$. All relevant side sizes are at least 20, above the inherited defect and span thresholds.

3 One effective-hole test for all high-hole states

The counted endpoint lemma of [1] says that if a positive prefix of size i has excess increment $y_i = t \geq 0$ and $i \geq 2t + 5$, its support, after scaling, is

$$[0, L]_{\mathbb{Z}} \setminus (H \cup (L - H)), \qquad H \subseteq [1, 2t]_{\mathbb{Z}}, \quad |H| \leq t.$$

It follows from the small-increment inverse lemma and Lev's dense subset-sum interval theorem as proved there. The following parameterized version makes the finite high-hole audit particularly short.

Lemma 3.1. *Let $C \geq 0$, $\beta \geq 1$ be integers and define*

$$K = \max\{k \geq 3 : k(k-3)/2 \leq C + 1\}, \quad D_* = C + K, \quad E_* = D_* + 1,$$

$$M = \max\left\{2\beta, 1 + \left\lfloor \frac{1 + \sqrt{9 + 8(E_* + 6\beta)}}{2} \right\rfloor\right\}, \quad J = \max\{K + 1, 2\beta + 5, M\}.$$

Suppose

$$\beta(r - J + 1) > D_*, \qquad T_{J-1} - 1 > D_*. \tag{5}$$

If a positive r-element set has, on a specified lattice, $2\beta < h < |\,\mathrm{FS}\,| - 1$ hull holes, then $\mu \geq r + C + 1$.

Proof. Assume $\mu \leq r + C$ and divide by the specified lattice step. The set is integral and primitive, since a gcd at least two would create at least $|\,\mathrm{FS}\,| - 1$ holes. Its support is not a progression. The last progression prefix of size $k \geq 3$, if there is one, is followed by a disjoint translate. Thus $\mu \geq r - 1 + k(k-3)/2$ and $k \leq K$. If there is no such prefix, use the tail starting at size four. In either case all increments in a tail starting no later than $K + 1$ obey

$$y_j \geq 1, \qquad \sum (y_j - 1) \leq D_*.$$

The first test forces an $i \in [J, r]$ with $1 \le y_i \le \beta$. The counted endpoint lemma gives a scaled support with $H \subseteq [1, 2\beta]$ and $|H| \le \beta$. Its primitive coefficient set makes its scale an integer. If the scale exceeded one, the first later off-scale element would give

$$y_j - 1 \ge T_{j-1} - j \ge T_{i-1} - 1 \ge T_{J-1} - 1 > D_*,$$

a contradiction. The scale is therefore one.

Every later increment excess is at most E_*. At a prefix of size $v \ge M$, its total $L \ge T_v$ and its next element $x > v \ge 2\beta$. If $x \ge L - 4\beta$, the new translate contributes at least $L - 6\beta$ points above L. This exceeds $v + 1 + E_*$, since the definition of M gives $T_M - 6\beta > M + 1 + E_*$ and the difference increases with v. Hence $2\beta < x < L - 4\beta$. The two central intervals overlap or abut; the lower holes remain H, and symmetry gives the upper holes. Induction leaves at most 2β holes, the required contradiction. □

Here is the complete finite rule used by the verifier. For $20 \le r \le 64$, $7 \le \beta \le 20$, $0 \le C \le 600$, retain the rows satisfying (5). The largest valid C for a pair (r, β) is only a convenient witness, not an asserted global optimum. For each $z \ge 15$, use only rows with $2\beta < z$. Recheck the chosen row exactly, and put

$$u_r = \max\big(\{r - 3\} \cup \{r + C + 1 : \text{eligible rows}\}\big), \qquad v_r = \max\{r - 14, u_r + 2 - 2r\}.$$

Equation (3) licenses the test for both sides. The second bound uses $\delta_0 \le r$. Equation (1) implies

$$c \le c_{\max} := \Omega - v_a - v_b - \max\{z, G - 2z, t_0\}. \tag{6}$$

Also $0 \le z \le \kappa \le \mathcal{T} \le \Omega$, so this is a finite enumeration with a proved upper limit. A negative $c_{\max}$ discards a state.

4 Coverage inequalities and all-active cases

Write $R_p = \max P$, $R_q = \max Q$, $M_p = \max D$, $M_q = -\min D$. Let S_p, S_q be the sums of the positive and negative magnitudes of D, and put

$$\varepsilon_p = \sum P - T_a = \mu(P) + h_P, \quad \varepsilon_q = \sum Q - T_b = \mu(Q) + h_Q,$$

$$F = S_p + S_q + c - 2(M_p + M_q) - 2(R_p + R_q).$$

Only the ten positive-positive and ten negative-negative pairs of the four peeled elements of each sign can lie outside $[-M_q - R_q, M_p + R_p]$. Thus

$$|2D| \le M_p + M_q + R_p + R_q + 21, \qquad F \le 18d + 126 + w \le 18d + 126 + 4z. \tag{7}$$

The outer magnitudes are denoted $p_1 > p_2 > p_3 > p_4 > M_p$ and $q_1 > q_2 > q_3 > q_4 > M_q$. Define

$$\lambda_p = p_1 - p_2 - p_3 - p_4, \quad \alpha_p = p_2 - p_3 - p_4, \quad \beta_p = 2p_2 - p_1 - p_3 - p_4,$$

and their reflected counterparts. The ten ordinary centers on each side are indexed as

$$v^p = (\lambda_p + p_1, \dots, \lambda_p + p_4, \lambda_p - q_1, \dots, \lambda_p - q_4, \alpha_p, \beta_p),$$
$$v^q = (\lambda_q + q_1, \dots, \lambda_q + q_4, \lambda_q - p_1, \dots, \lambda_q - p_4, \alpha_q, \beta_q).$$

Indices are $0, \dots, 9$. The first paired unit consists of $p4, q4$. In branch A, $\beta_p + \beta_q \le 0$, the second is $p9, q9$. In branch B, $\beta_p + \beta_q > 0$, the second is $p5, q5$. The other sixteen units are single centers. Each of the eighteen units has capacity d in the inherited cover; an empty unit costs d in c. Losses are counted by formal slots even if numerical centers coincide.

If all units are nonempty, the active-center proof in [1] applies, regardless of whether $c < d$. After a possible reflection it gives one of

$$F \geq F_A(a,b) + \tfrac{1}{2}\varepsilon_p + \varepsilon_q, \qquad F_A = \tfrac{1}{2}T_a + T_b + 6a + 8b + 85, \tag{8}$$

$$F \geq F_B(a,b) + \tfrac{2}{3}\varepsilon_p + \varepsilon_q, \qquad F_B = \tfrac{2}{3}T_a + T_b + 6a + \tfrac{22b+236}{3}. \tag{9}$$

The proof absorbs all out-of-hull corrections into the available spans. Its hypothesis is actual activity; $c < d$ is a sufficient condition for it.

For $0 \leq z \leq 14$, use $u_a = u_b = 0$, $v_a = v_b = 0$ at $z = 0$; otherwise use $u_a = a - 3$, $u_b = b - 3$, $v_a = a - 14$, $v_b = b - 14$. For $z \geq 15$, use Section 3. In both cases $\varepsilon_p \geq u_a + z$, $\varepsilon_q \geq u_b + z$ (at zero holes this is simply nonnegativity). Subtracting (7) from both (8) and (9) gives the checked margins:

n	Low states	Low margin	High states	High margin	High $c_{\max}$	Empty-eligible low states
103	261	215/2	0	–	–	9
102	300	155/2	32	365	6	66
101	333	113/2	399	248	36	165

States are necessary scalar relaxations, not constructed sets. All margins are strictly positive. Every surviving high-hole state has $c_{\max} < d$, so all high-hole cases are excluded. Every low-hole case with no empty unit is excluded as well. Only the empty-eligible low states remain.

5 Exactly one empty unit and at most one outside center

For low holes the basic bound is

$$c \leq \Omega - G \quad (z = 0), \qquad c \leq \Omega - G + 2z - (m - 28) \quad (1 \leq z \leq 14).$$

At the three sizes it is always less than $2d$, hence an empty-unit case has exactly one such unit. Put $\delta = c - d \geq 0$.

5.1 Outer integrality and exhaustive contributor choices

Every contributing single center is integral, since it translates the integral set D into the integral target support. If all four own centers of a side remain, the identity $\lambda_p = -v_0^p + v_1^p + v_2^p + v_3^p$ gives integrality of that side. If one own center on the positive side is missing, all negative own centers remain. Thus all q_i are integral. The always unpaired positive centers $\lambda_p - q_3, \lambda_p - q_4$ remain and force λ_p integral. The other three p_i and the defining identity force the last p_i integral. Reflection covers the other case. All outer strict gaps are therefore at least one.

The first pair always has at most one contributing member: its center sum is less than $-(M_p + M_q)$. If present, reflect to retain $p4$. In branch B the second pair also has at most one contributor. In branch A this holds if both unpaired centers $p5, q5$ remain, because

$$\beta_p + \beta_q = -(v_5^p + v_5^q) - 2(p_3 + p_4 + q_3 + q_4) < -(M_p + M_q).$$

If $p5$ or $q5$ is the missing unit in branch A, omit the entire second pair from the retained list. Allowing it to split only weakens the model.

The complete enumeration is now explicit. Choose a branch and a missing unit among the two pairs and sixteen singles. Retain all other singles. Retain $p4$ unless the first pair is missing. For the second pair retain each of its two possible contributors in a separate mode, except when that pair is missing or deliberately omitted. There are 33 modes in A and 35 in B. Sixty-six modes retain 17 centers; the two split-pair modes retain 16. If the first pair is missing both choices for the second are still included. This loses no configuration.

5.2 Hull losses and the five regimes

For a retained positive center v, let l_v, u_v be the counts of $v + D$ below 1 and above S_p. Reflect for a negative center. Then

$$\sum_{v \text{ retained}} (l_v + u_v) \le \delta. \tag{10}$$

Within its hull a positive center satisfies

$$-v - l_v \le -(b+5), \qquad v - u_v - S_p \le -(a+4). \tag{11}$$

Outside, the valid replacements are

$$v < 0: \quad -v - l_v - M_p \le -d, \quad l_v \ge b + 5, \tag{12}$$

$$v > S_p: \quad v - u_v - S_p - M_q \le -(d-1), \quad u_v \ge a + 5. \tag{13}$$

These follow by counting integer points in the occupied positive and negative spans; equivalently substitute $U = M_p - a - 4$, $V = M_q - b - 4$ in the global quantile bounds of [1]. The opposite endpoint inequality in (11) remains valid. Thus each outside center costs at least $\min(a, b) + 5$.

For 103, write $a, b = 43 \mp k$ up to reflection. An empty unit gives $\delta \le 21 - k^2$, $k \le 4$, and $z = 0$; all retained centers are in-hull. For 102, write $a, b = 42 - k, 43 + k$. At zero holes $\delta \le 45 - k(k+1)$, $k \le 6$, and this is less than $47 - k$. At positive low holes $\delta \le 16 - k(k+1)$, an even smaller bound. All retained centers are again in-hull.

For 101, write $a, b = 42 \mp k$. At positive low holes, $\delta \le 41 - k^2 < 47 - k$ and $k \le 6$. The core masses satisfy $\varepsilon_p \ge a - 2$, $\varepsilon_q \ge b - 2$. At zero holes, $\delta \le 69 - k^2$, $k \le 8$. Since

$$2(47 - k) - (69 - k^2) = (k-1)^2 + 24 > 0,$$

at most one retained center is outside its hull. If one is outside, $k^2 - k \le 22$, hence $k \le 5$ and $37 \le a, b \le 47$.

All remaining configurations are covered by these relaxed regimes, with $a = a_0 + t$, $b = b_0 - t$, $0 \le t \le t_{\max}$:

Regime	n	a_0	b_0	$t_{\max}$	$\delta_{\max}$	Upper F	Modes	Retained centers
N103	103	39	47	8	21	1836	68	All in-hull
N102	102	36	49	13	45	1874	68	All in-hull
Z	101	34	50	16	69	1800	68	All in-hull
P	101	36	48	12	41	1856	68	All in-hull; mass bonus
E	101	37	47	10	69	1800	2308	One designated exception

Regime E enumerates both endpoints of every retained center in every mode, giving $2(66 \cdot 17 + 2 \cdot 16) = 2308$ systems. No guessed exception location is used. The scalar verifier also checks every state against these regime ranges and strict outside-loss thresholds.

6 Complete linear systems and exact certificates

All variables

$$x = (p_1, \dots, p_4, q_1, \dots, q_4, M_p, M_q, R_p, R_q, S_p, S_q, \delta, t, (l_v, u_v)_{v \text{ retained}})$$

are nonnegative real numbers. We deliberately relax t and all losses to real variables. Use the following rows, all oriented $Ax \le b$:

Row	Constraint (and reflected version)
D_i^p $(i=1,2,3)$	$p_{i+1}-p_i \le -1$
D_4^p	$M_p - p_4 \le -1$
E_p	$R_p - M_p \le -4$
G_p, G_q	$-R_p + t \le -a_0$, $-R_q - t \le -b_0$
H_p	$2M_p + 6R_p - 2S_p + s_0 t \le -(s_0 a_0 - j_0 + 12)$
H_q	$2M_q + 6R_q - 2S_q - s_0 t \le -(s_0 b_0 - j_0 + 12)$
cap, range	$\delta \le \delta_{\max}$, $t \le t_{\max}$
branch	$\beta_p + \beta_q \le 0$ (A), $-\beta_p - \beta_q \le -1$ (B)
L_v, U_v	Equation (11), substituting $a = a_0 + t$, $b = b_0 - t$
loss	Equation (10)
Fupper	$S_p + S_q + \delta - 2(M_p + M_q + R_p + R_q) \le F_{\max} - d$

For N103 use $(s_0, j_0) = (87, 1849)$; for N102, Z, E use $(85, 1764)$; for P use $(87, 1768)$. The mass rows follow from

$$S_p \ge T_a + \varepsilon_p + 3R_p + 6 + M_p$$

and its reflection, together with the exact tangent inequalities

$$2T_r - (87r - 1849) = (r-43)^2 \ge 0, \qquad 2T_r - (85r - 1764) = (r-42)^2 \ge 0.$$

In P add $2\varepsilon_p \ge 2a - 4$ and its reflection, giving $2(T_r + r - 2) \ge 87r - 1768$. In E replace only the designated L_v or U_v by (12) or (13), and add its unavoidable loss row. The opposite endpoint row is unchanged. It is unnecessary to impose the actual outside location, because omitting it enlarges the model.

Lemma 6.1 (Integer contradiction rule)**.** *If $x \ge 0$, $Ax \le b$ and a nonnegative integer row vector w satisfies*

$$wA \ge 0 \text{ coefficientwise}, \qquad wb < 0,$$

the system is inconsistent.

Proof. One would have $0 \le wAx \le wb < 0$. □

The file `data/upper100_certificates.json` contains 189 distinct integer row-weight dictionaries and an assignment for every one of the 2,580 model keys. Row labels are exactly those in `verification/upper100.py`; for example, `Lp5` denotes the lower row for the positive center with index 5, and `outside` denotes the extra unavoidable-loss row in E. The verifier reconstructs all models from the rules above, requires exact equality between the generated key set and the assigned key set, checks that every referenced row exists, and checks Lemma 6.1 using integers. Every system passes. The minimum scale-independent contradiction gap $-wb/\sum_i w_i$ is $8/101 > 0$.

These are finite proof data, not optimizer verdicts. The optional discovery script uses SciPy to propose weights, converts them to rational numbers and then integers, and refuses to save any vector that fails the exact rule. Neither SciPy nor floating-point arithmetic is used by the verification script. The freshly generated vectors need not agree with the archived multipliers.

7 Closing the descent and reproducing the result

Proof of Theorem 1.1. Start with $N \le 103$ from [1]. A 103-element feasible set must contain zero, since otherwise adjoining zero contradicts that bound. Integralize and use the checked lattice gates. Sections 3–4 exclude all high-hole states and all low-hole states with no empty unit. The remaining states have one empty unit and belong to N103, whose 68 exact linear contradictions exclude them. Thus $N \le 102$.

At size 102 the same argument applies, and the N102 systems exclude the remaining empty-unit states. Thus $N \le 101$. At size 101, the remaining states are partitioned into positive low holes

(P), zero holes with all retained centers in-hull (Z), and zero holes with one retained exception (E). All their systems are inconsistent. Therefore $N \leq 100$. At no stage is feasibility assumed to pass to arbitrary subsets. □

From the repository root, the verification commands are

```
python -I -S -B verification/upper103.py --output .work/base103.json
python -I -S -B verification/upper100.py --output .work/upper100.json
```

Python 3.10 or later suffices; no third-party package or network is needed. The new complete result and receipt are in `results/2026-10-16/upper100/`. A second run with `-O` produces the identical JSON. Checks use explicit exceptions, not Python assertions.

Seven negative controls reject an invalid hole witness, zero weights, a noncontradictory positive certificate, a negative weight, an absent row, a missing case, and an attempt to certify cardinality 100 using this three-endpoint routine. Hashes in the output bind the run to the verifier and certificate bytes. These checks do not replace the written proof that actual configurations satisfy the rows, nor do they formally certify the external structural theorems inherited from [1].

Optional regeneration, requiring SciPy, is

```
python experiments/generate_upper100_certificates.py \
  --output .work/regenerated_upper100.json
python verification/upper100.py \
  --certificates .work/regenerated_upper100.json
```

The next endpoint has a concrete obstruction to the current one-empty argument. At $n = 100$, $a = 41$, $b = 42$, $z = 0$, one has

$$d = 92, \quad \Omega = 356, \quad G = 172, \quad c \leq 184 = 2d.$$

Two empty units are therefore allowed by this relaxation. Excluding them, or strengthening the budget to remove them, is a necessary next step. This is an obstruction in the proof method, not a feasible 100-element set.

References

[1] Sum-of-Subset project, *Subset Sums in a Fourfold Sumset: A checked continuation from the upper bound 115 to 103*, 14 October 2026. `reports/upper103_from115_20261014.tex`. Its Sections 1–6 specify the audited foundation, external theorems, and structural lemmas used here.

[2] Sum-of-Subset project, *Complete upper-86 report*, 5 October 2026. `reports/upper86_complete_20261005.tex`, sections on upper bounds 102 and 100 and their certificate appendices. Source of the empty-unit strategy; its complete endpoint chain is outside this audit.